

AI Automation Proposal

Maintenance Intake Triage Assistant for Summit Ridge DataHub

Assumptions

I assumed Summit Ridge earns recurring management fees and spends a lot of staff time turning owner messages, board requests, and vendor updates into tracked work. I also assumed the DataHub is meant to become the trusted source for HOA context, board contacts, vendor relationships, and compliance notes.

Problem

Maintenance intake is repetitive and easy to route incorrectly. A coordinator has to identify the HOA, understand the issue, decide the trade, check vendor fit, notice missing information, and write a response. SQL is useful after the facts are known, but it does not read messy emails or photos well.

What I would build

I would build an intake assistant that reads incoming owner emails, web form entries, and vendor notes. It would produce a draft ticket with the association, issue type, urgency, missing details, suggested vendor choices, COI warnings, and a short draft reply for staff to approve.

How I would build it

The AI part would classify and summarize the messy message. The normal code and database queries would handle the reliable checks, such as which vendors serve the association, whether a COI is on file, and which fields are missing. Every draft would be stored with the source message, model output, confidence, and human approval status.

Guardrails

A human would approve anything sent to owners, boards, or vendors. Low confidence matches, emergency language, legal topics, and missing HOA matches would go to a review queue. Vendor suggestions would show why they were selected and would clearly warn when COI information is missing or not on file.

Why this is worth doing

This would save time on a task that happens every day, while still keeping people in control. It uses AI where AI is useful, reading messy language and drafting a clear summary, and uses deterministic database checks where accuracy matters.

AI use on this proposal

I used AI to help draft and tighten this proposal, then I edited it to match the case study, my schema, and the kind of workflow I would actually build next.